

Make A Smart Presentation IR Pointer Using Arduino

Pamarthi Kanakaraja

Smart presentation pointers are readily available in the market but are expensive. Here is a simple, easy-to-build, inexpensive and infrared (IR)-based smart slide pointer using Arduino Leonardo.

Slide presentations using PowerPoint can be frustrating because you may need to click the mouse to navigate. This IR remote-based slide pointer looks ergonomic when compared to a mouse or arrow keys on the keyboard. The block diagram of the smart slide pointer is shown in Fig. 1. The author's prototype is shown in Fig. 2.

Circuit and working

The circuit diagram of the smart slide pointer is shown in Fig. 3. It is built around Arduino Leonardo (Board1), IR sensor TSOP1738 (IRRX1) and a remote control.

Connections are simple. Connect GND pin of the sensor to GND pin, Vcc pin to 5V pin, and O/P pin to digital pin 2 of Arduino Leonardo, respectively, as shown in Fig. 3.

Arduino Leonardo. This board is based on ATmega32u4 microcontroller (MCU). It has twenty digital input/output (I/O) pins (of which seven can be used as PWM outputs and twelve as analogue inputs), 16MHz crystal oscillator, micro-USB connection, power jack, ICSP header and reset button.

It contains everything needed to support the MCU; simply connect it to a computer with a USB cable or power it with an AC-to-DC adaptor or battery to get started. Arduino Leonardo dif-

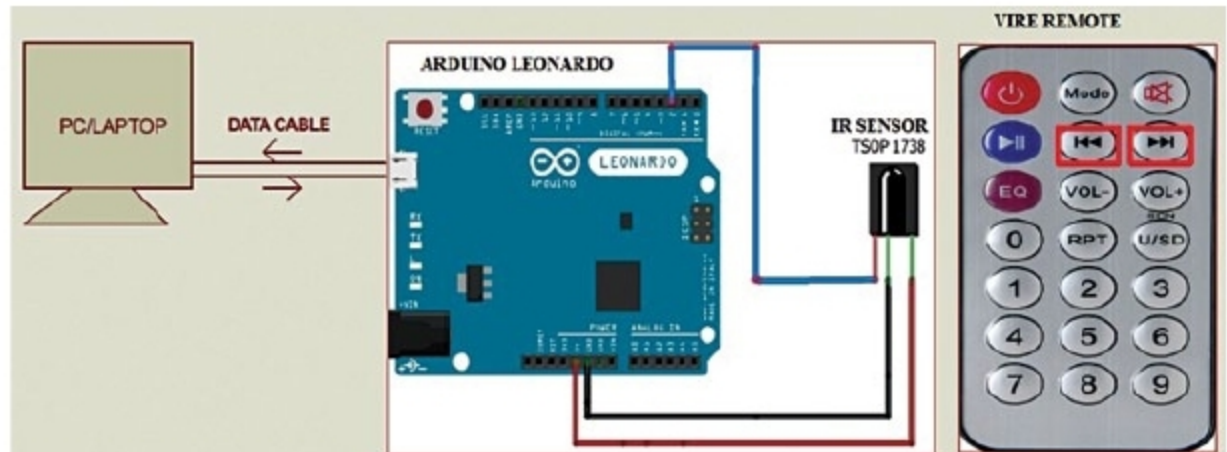


Fig. 1: Block diagram of the smart slide pointer

fers from all preceding boards—in this board ATmega32U4 has inbuilt USB communication, eliminating the need for a secondary processor.

The software

Circuit operation is done through the software program loaded into the internal memory of Arduino Leonardo. The program/sketch is written in Arduino programming language. Arduino IDE 1.6.4 is used to compile and upload the program. ATmega32U4 on Arduino Leonardo board comes with a pre-programmed boot loader that allows you to upload a new code to it without using an external hardware programmer.

The sketch is at the heart of the system and carries out all major functions. Program coding is divided into two parts. In the first part, the program decodes the keys/buttons that you want to use from your TV/MP3 remote. You will be able to decode the IR signal transmitted by each button from the remote.

In the second part, each decoded signal obtained from the first part is assigned to a particular function to be used in the slide pointer application.

In the first part, IR_decoder.ino code is used. Whereas, in the second part, ppt_pointer.ino code is used.



Fig. 2: Author's prototype

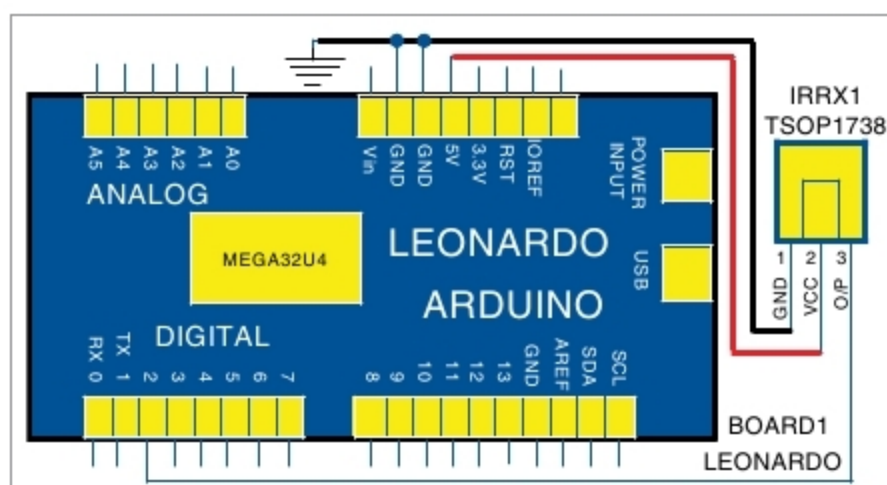


Fig. 3: Circuit diagram of the smart slide pointer using Arduino Leonardo

EFY Note

The source code of this project is available for free download at source.efymag.com